Systems Analysis and Control

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Lecture 7: The Partial Fraction Expansion

Introduction

In this Lecture, you will learn: Transfer Functions and Form of the Output

The Partial Fraction Expansion

- How poles relate to dominant modes
- Expansion using single poles
- Repeated Poles
- Complex Pairs of Poles
  - Inverse Laplace

Will help us to understand metrics of response

Rational Transfer Functions

Theorem 1.

Any bounded, linear, causal, time-invariant system, G, has a **Transfer Function**, $\hat{G}$, *so that if* $y = Gu$, *then*

$$\hat{y}(s) = \hat{G}(s)\hat{u}(s)$$

Important! The transfer functions are alway rational.

Definition 2.

A **Rational Transfer Function** is the ratio of two polynomials:

$$\hat{G}(s) = \frac{n(s)}{d(s)}$$

Definition 3.

The real or *complex* number s_p is a **Pole** of $\hat{G}(s) = \frac{n(s)}{d(s)}$ if $d(s_p) = 0$.

We can factor the denominator using the poles: $\frac{n(s)}{d(s)} = \frac{n(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)}$

The Output will be Rational

	$u(t)$	$\hat{u}(s)$
Step	$\mathbf{1}(t)$	$\frac{1}{s}$
Power	t^m	$\frac{m!}{s^{m+1}}$
Exp.	e^{-at}	$\frac{1}{s+a}$
Power	$\frac{t^{m-1}e^{-at}}{(m-1)!}$	$\frac{1}{(s+a)^m}$
Sine	$\sin(at)$	$\frac{a}{s^2+a^2}$
Cosine	$\cos(at)$	$\frac{s}{s^2+a^2}$
Impulse	$\delta(t)$	1

Note: For all our simple inputs, $\hat{u}(s)$ is *rational*.

Now, combine with

$$\hat{y}(s) = \hat{G}(s)\hat{u}(s)$$

and $\hat{G}(s) = \frac{n(s)}{d(s)}$

Conclusion: The output $\hat{y}(s)$ is also going to be *rational*

Question: What does this say about $y(t) = \mathcal{L}^{-1}(\hat{y}(s))$?

Problem: Calculating the Inverse Laplace Transform can be tricky. e.g.

$$\hat{y}(s) = \frac{s^3 + s^2 + 2s - 1}{s^4 + 3s^3 - 2s^2 + s + 1}$$

Answer: Use the Partial Fraction Expansion

Partial Fraction Expansion

Strictly Proper Rational Functions

Definition 4.

The **Degree** of a polynomial $n(s)$, is the highest power of s with a nonzero coefficient.

Example: The degree of $n(s)$ is 4

$$n(s) = s^4 + .5s^2 + 1$$

Important: For Partial Fraction Expansion to work, $\hat{y}$ must be *Strictly Proper*

Definition 5.

A rational function $\hat{y}(s) = \frac{n(s)}{d(s)}$ is **Strictly Proper** if the degree of $n(s)$ is less than the degree of $d(s)$.

- We assume that $n(s)$ has lower degree than $d(s)$
- Otherwise, perform long division until we have a **strictly proper** remainder

$$\frac{s^3 + 2s^2 + 6s + 7}{s^2 + s + 5} = s + 1 + \frac{2}{s^2 + s + 5}$$

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$$\frac{s^3 + 2s^2 + 6s + 7}{s^2 + s + 5} = s + 1 + \frac{2}{s^2 + s + 5}$$

Example of polynomial long division:

$$y(x) = \frac{3x^3 + x^2 + 2x + 5}{x^2 + 2x + 1}$$

$$\begin{array}{r|l} & 3x \quad -5 \\ \hline x^2 + 2x + 1 & \big) \, 3x^3 \quad +x^2 \quad +2x \quad +5 \\ & - \\ & \underline{\;3x^3 \quad +6x^2 \quad +3x\;\;\;\;\;\;} \\ & \qquad -5x^2 \quad -x \quad +5 \\ & \qquad - \\ & \underline{\qquad -5x^2 \quad -10x \quad -5} \\ & \qquad\qquad\quad 9x \quad +10 \end{array}$$

So that

$$y(x) = \frac{3x^3 + x^2 + 2x + 5}{x^2 + 2x + 1} = 3x - 5 + \frac{9x + 10}{x^2 + 2x + 1}$$

Poles and Inverse Laplace Transforms

A Strictly Proper Rational Function is The Sum of Poles: We can *usually* find coefficients r_i such that

$$\frac{n(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)} = \frac{r_1}{s-p_1} + \ldots + \frac{r_n}{s-p_n}$$

- Except in the case of repeated poles.

Poles Dominate the Motion: Because a signal is the sum of poles, The inverse Laplace has the form

$$u(t) = r_1 e^{p_1 t} + \ldots + r_n e^{p_n t}$$

- p_i may be complex.
  - If p_i are complex, r_i may be complex.
- Doesn't hold for repeated poles.

Examples we have seen before

Simple State-Space: Step Response

$$\hat{y}(s) = \frac{s-1}{(s+1)s} = \frac{2}{s+1} - \frac{1}{s}$$

$$y(t) = e^{-t} - \frac{1}{2}\mathbf{1}(t)$$

Suspension System: Impulse Response

$$\hat{y}(s) = \frac{1}{J}\frac{1}{s^2 - \frac{Mgl}{2J}} = \frac{1}{J}\sqrt{\frac{2J}{Mgl}}\left(\frac{1}{s - \sqrt{\frac{Mgl}{2J}}} - \frac{1}{s + \sqrt{\frac{Mgl}{2J}}}\right)$$

$$y(t) = \frac{1}{J}\sqrt{\frac{2J}{Mgl}}\left(e^{\sqrt{\frac{Mgl}{2J}}t} - e^{-\sqrt{\frac{Mgl}{2J}}t}\right)$$

Simple State-Space: Sinusoid Response

$$\hat{y}(s) = \frac{s-1}{(s+1)(s^2+1)} = \left(\frac{s}{s^2+1} - \frac{1}{s+1}\right)$$

$$y(t) = \frac{1}{2}\cos t - \frac{1}{2}e^{-t}$$

Partial Fraction Expansion

Conclusion: If we can find coefficients r_i such that

$$\begin{aligned}\hat{u}(s) = \frac{n(s)}{d(s)} &= \frac{n(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)} \\ &= \frac{r_1}{s-p_1} + \cdots + \frac{r_n}{s-p_n},\end{aligned}$$

then this is the **PARTIAL FRACTION EXPANSION** of $\hat{u}(s)$.

We will address several cases of increasing complexity:

1. $d(s)$ has all real, non-repeating roots.
2. $d(s)$ has all real roots, some repeating.
3. $d(s)$ has complex, repeated roots.

Case 1: Real, Non-repeated Roots

This case is the easiest:

$$\hat{y}(s) = \frac{n(s)}{(s-p_1)\cdots(s-p_n)}$$

where p_n are all real and distinct.

Theorem 6.

For case 1, $\hat{y}(s)$ can always be written as

$$\hat{y}(s) = \frac{r_1}{s-p_1} + \cdots + \frac{r_n}{s-p_n}$$

where the r_i are all real constants

The trick is to solve for the r_i

- There are n unknowns - the r_i.
- To solve for the r_i, we evaluate the equation for values of s.
  - Potentially unlimited equations.
  - We only need n equations.
  - We will evaluate at the points $s = p_i$.

Case 1: Real, Non-repeated Roots

Solving

We want to solve

$$\hat{y}(s) = \frac{r_1}{s-p_1} + \cdots + \frac{r_n}{s-p_n} \qquad \text{for } r_1, r_2, \ldots$$

For each r_i, we Multiply by $(s-p_i)$.

$$\begin{aligned}\hat{y}(s)(s-p_i) &= r_1\frac{s-p_i}{s-p_1} + \cdots + r_i\frac{s-p_i}{s-p_i} + \cdots + r_n\frac{s-p_i}{s-p_n} \\ &= r_1\frac{s-p_i}{s-p_1} + \cdots + r_i + \cdots + r_n\frac{s-p_i}{s-p_n}\end{aligned}$$

Evaluate the Right Hand Side at $s = p_i$: We get

$$\begin{aligned}&r_1\frac{p_i-p_i}{p_i-p_1} + \cdots + r_i + \cdots + r_n\frac{p_i-p_i}{p_i-p_n} \\ &= r_1\frac{0}{p_i-p_1} + \cdots + r_i + \cdots + r_n\frac{0}{p_i-p_n} = r_i\end{aligned}$$

Setting $LHS = RHS$, we get a simple formula for r_i:

$$r_i = \hat{y}(s)(s-p_i)\big|_{s=p_i}$$

Case 1: Real, Non-repeated Roots

Calculating r_i

So we can find all the coefficients:

$$r_i = \hat{y}(s)(s-p_i)|_{s=p_i}$$

$$\begin{aligned}\hat{y}(s)(s-p_i) &= \frac{n(s)}{(s-p_1)\cdots(s-p_{i-1})(s-p_i)(s-p_{i+1})\cdots(s-p_n)}(s-p_i) \\ &= \frac{n(s)}{(s-p_1)\cdots(s-p_{i-1})(s-p_{i+1})\cdots(s-p_n)}\end{aligned}$$

- This is $\hat{y}$ with the pole $s-p_i$ removed.

Definition 7.

The **Residue** of $\hat{y}$ at $s=p_i$ is the value of $\hat{y}(p_i)$ with the pole $s-p_i$ removed.

Thus we have

$$r_i = \hat{y}(s)(s-p_i)|_{s=p_i} = \frac{n(p_i)}{(p_i-p_1)\cdots(p_i-p_{i-1})(p_i-p_{i+1})\cdots(p_i-p_n)}$$

Example: Case 1

$$\hat{y}(s) = \frac{2s+6}{(s+1)(s+2)}$$

We first separate $\hat{y}$ into parts:

$$\hat{y}(s) = \frac{2s+6}{(s+1)(s+2)} = \frac{r_1}{s+1} + \frac{r_2}{s+2}$$

Residue 1: we calculate:

$$r_1 = \frac{2s+6}{(s+1)(s+2)}(s+1)|_{s=-1} = \frac{2s+6}{s+2}|_{s=-1} = \frac{-2+6}{-1+2} = \frac{4}{1} = 4$$

Residue 2: we calculate:

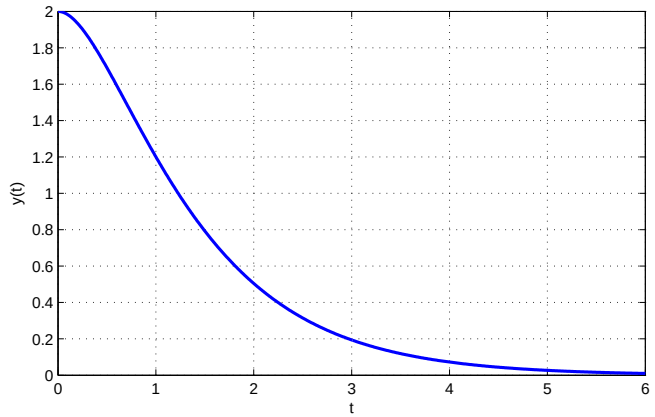
$$r_2 = \frac{2s+6}{(s+1)(s+2)}(s+2)|_{s=-2} = \frac{2s+6}{s+1}|_{s=-2} = \frac{-4+6}{-2+1} = \frac{2}{-1} = -2$$

Thus

$$\hat{y}(s) = \frac{4}{s+1} + \frac{-2}{s+2}$$

Concluding,

$$y(t) = 4e^{-t} - 2e^{-2t}$$



Case 2: Real, Repeated Roots

Sometimes $\hat{y}$ has a repeated pole:

$$\hat{y}(s) = \frac{n(s)}{(s-p_1)^q(s-p_2)\cdots(s-p_n)}$$

In this case, we **CAN NOT** use simple expansion (As in case 1). Instead we have

$$\hat{y}(s) = \left(\frac{K_q}{(s-p_1)} + \frac{K_{q-1}}{(s-p_1)^2} + \cdots + \frac{K_1}{(s-p_1)^q}\right) + \frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n}$$

- The K_i are still real-valued.

Multiple Repeated Poles: Find the K_i and L_i

$$\frac{(s+3)^2}{(s+2)^3(s+1)^2} = \frac{K_3}{s+2} + \frac{K_2}{(s+2)^2} + \frac{K_1}{(s+2)^3} + \frac{L_2}{s+1} + \frac{L_1}{(s+1)^2}$$

Case 2: Real, Repeated Roots

Problem: We have more coefficients to find.

- q coefficients for each repeated root $(s - p_i)^q$.

First Step: Solve for $r_2, \cdots, r_n$ as before:
If p_i is not a repeated root, then

$$r_i = \frac{n(p_i)}{(p_i - p_1)^q \cdots (p_i - p_{i-1})(p_i - p_{i+1}) \cdots (p_i - p_n)}$$

Case 2: Real, Repeated Roots

New Step: Multiply by $(s-p_1)^q$ to get the coefficient K_q.

$$\begin{aligned}\hat{y}(s)(s-p_1)^q &= \left(K_q\frac{(s-p_1)^q}{(s-p_1)} + K_{q-1}\frac{(s-p_1)^q}{(s-p_1)^2} + \cdots + K_1\frac{(s-p_1)^q}{(s-p_1)^q}\right) \\ &\quad + r_2\frac{(s-p_1)^q}{s-p_2} + \cdots + r_n\frac{(s-p_1)^q}{s-p_n} \\ &= \left(K_q(s-p_1)^{q-1} + K_{q-1}(s-p_1)^{q-2} + \cdots + K_1\right) \\ &\quad + r_2\frac{(s-p_1)^q}{s-p_2} + \cdots + r_n\frac{(s-p_1)^q}{s-p_n}\end{aligned}$$

and evaluate at the point $s=p_i$ to get .

$$K_1 = \hat{y}(s)(s-p_1)^q\big|_{s=p_1}$$

- K_1 is $\hat{y}(p_1)$ with the repeated pole removed.

Case 2: Real, Repeated Roots

To find the remaining coefficients, we Differentiate:

$$\frac{d}{ds}\left(\hat{y}(s)(s-p_1)^q\right) = \frac{d}{ds}\Bigg(\left(K_q(s-p_1)^{q-1} + K_{q-1}(s-p_1)^{q-2} + \cdots + K_1\right) + r_2\frac{(s-p_1)^q}{s-p_2} + \cdots + r_n\frac{(s-p_1)^q}{s-p_n}\Bigg)$$

to get

$$\begin{aligned}\frac{d}{ds}&\left(\hat{y}(s)(s-p_1)^q\right)\\ &= \left((q-1)K_q(s-p_1)^{q-2} + (q-2)K_{q-1}(s-p_1)^{q-3} + \cdots + K_2\right)\\ &\quad + \frac{d}{ds}(s-p_1)^q\left(\frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n}\right)\end{aligned}$$

Evaluating at the point $s = p_1$.

$$K_2 = \frac{d}{ds}\left(\hat{y}(s)(s-p_1)^q\right)\Big|_{s=p_1}$$

Case 2: Real, Repeated Roots

If we differentiate again, we get

$$\begin{aligned}\frac{d^2}{ds^2}\left(\hat{y}(s)(s-p_1)^q\right) \\ &= \left((q-1)(q-2)K_q(s-p_1)^{q-3} + \cdots + 2K_3\right) \\ &\quad + \frac{d^2}{ds^2}(s-p_1)^q\left(\frac{r_2}{s-p_2} + \cdots + \frac{r_n}{s-p_n}\right)\end{aligned}$$

Evaluating at $s-p_1$, we get

$$K_3 = \frac{1}{2}\left[\frac{d^2}{ds^2}\hat{y}(s)(s-p_1)^q\right]\Big|_{s=p_1}$$

Extending this indefinitely for $j = 1, \cdots, q$

$$K_j = \frac{1}{(j-1)!}\left[\left(\frac{d}{ds}\right)^{j-1}\hat{y}(s)(s-p_1)^q\right]\Big|_{s=p_1}$$

Of course, calculating this derivative is often difficult.

Example: Real, Repeated Roots

Expand a simple example

$$\hat{y}(s) = \frac{s+3}{(s+2)^2(s+1)} = \frac{K_2}{s+2} + \frac{K_1}{(s+2)^2} + \frac{r_2}{s+1}$$

$$\hat{y}(s) = \frac{s-1}{(s+1)^2(s+3)} = \frac{K_2}{s+1} + \frac{K_1}{(s+1)^2} + \frac{r_2}{s+3}$$

First Step: Calculate r_2

$$r_2 = \frac{s+3}{(s+2)^2}\Big|_{s=-1} = \frac{-1+3}{(-1+2)^2} = \frac{2}{1} = 2$$

Second Step: Calculate r_{12}

$$K_1 = \frac{s+3}{s+1}\Big|_{s=-2} = \frac{-2+3}{-2+1} = \frac{1}{-1} = -1$$

The Difficult Step: Calculate r_{11}:

$$K_2 = \frac{d}{ds}\left(\frac{s+3}{s+1}\right)\Big|_{s=-2}$$

$$= \left(\frac{1}{s+1} - \frac{s+3}{(s+1)^2}\right)\Big|_{s=-2} = \frac{1}{-2+1} - \frac{-2+3}{(-2+1)^2} = \frac{1}{-1} - \frac{1}{1} = -2$$

Example: Real, Repeated Roots

So now we have

$$\hat{y}(s) = \frac{s+3}{(s+2)^2(s+1)} = \frac{-2}{s+2} + \frac{-1}{(s+2)^2} + \frac{2}{s+1}$$

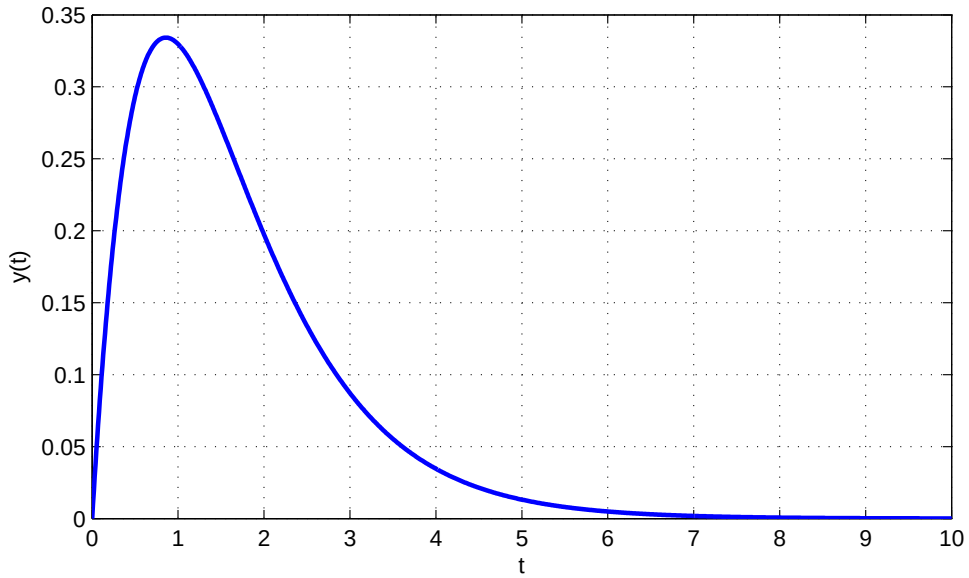
Question: what is the Inverse Fourier Transform of $\frac{1}{(s+2)^2}$?

Recall the Power Exponential:

$$\frac{1}{(s+a)^m} \rightarrow \frac{t^{m-1}e^{-at}}{(m-1)!}$$

Finally, we have:

$$y(t) = -2e^{-2t} - te^{-2t} + 2e^{-t}$$



Case 3: Complex Roots

Most signals have complex roots (More common than repeated roots).

$$\hat{y}(s) = \frac{3}{s(s^2 + 2s + 5)}$$

has roots at $s = 0$, $s = -1 - 2\imath$, and $s = -1 + 2\imath$.

Note that:

- Complex roots come in pairs.
- Simple partial fractions will work, but NOT RECOMMENDED
  - Coefficients will be complex.
  - Solutions will be complex exponentials.
  - Require conversion to real functions.

Best to Separate out the Complex pairs as:

$$\hat{y}(s) = \frac{n(s)}{(s^2 + as + b)(s - p_1) \cdots (s - p_n)}$$

Case 3: Complex Roots

Complex pairs have the expansion

$$\frac{n(s)}{(s^2+as+b)(s-p_1)\cdots(s-p_n)} = \frac{k_1 s + k_2}{s^2+as+b} + \frac{r_1}{s-p_1} + \cdots + \frac{r_n}{s-p_n}$$

Note that there are two coefficients for each pair: k_1 and k_2.

NOTE: There are several different methods for finding k_1 and k_2.

First Step: Solve for $r_2, \cdots, r_n$ as normal.

Second Step: Clear the denominator. Multiply equation by all poles.

$$n(s) = (s^2+as+b)(s-p_1)\cdots(s-p_n)\left(\frac{k_1 s + k_2}{s^2+as+b} + \frac{r_1}{s-p_1} + \cdots + \frac{r_n}{s-p_n}\right)$$

Third Step: Solve for k_1 and k_2 by examining the coefficients of powers of s.

Warning: May get complicated or impossible for multiple complex pairs.

Case 3 Example

Take the example

$$\hat{y}(s) = \frac{2(s+2)}{(s+1)(s^2+4)} = \frac{k_1 s + k_2}{s^2+4} + \frac{r_1}{s+1}$$

First Step: Find the simple coefficient r_1.

$$r_1 = \frac{2(s+2)}{s^2+4}|_{s=-1} = 2\frac{-1+2}{-1^2+4} = 2\frac{1}{5} = \frac{2}{5}$$

Next Step: Multiply through by $(s^2+4)(s+1)$.

$$2(s+2) = (s+1)(k_1 s + k_2) + r_1(s^2+4)$$

Expanding and using $r_1 = 2/5$ gives:

$$2s + 4 = (k_1 + 2/5)s^2 + (k_2 + k_1)s + k_2 + 8/5$$

Case 3 Example

Recall the equation

$$2s + 4 = (k_1 + 2/5)s^2 + (k_2 + k_1)s + k_2 + 8/5$$

Equating coefficients gives 3 equations:

- s^2 **term** - $0 = k_1 + 2/5$
- s **term** - $2 = k_2 + k_1$
- s^0 **term** - $4 = k_2 + 8/5$

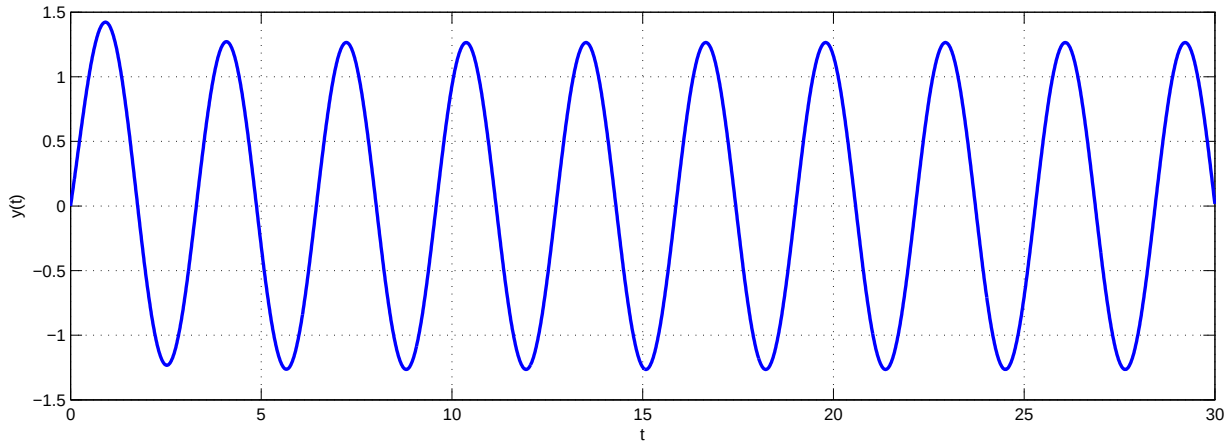
An over-determined system of equations (but consistent)

- First term gives $k_1 = -2/5$
- Second term gives $k_2 = 2 - k_1 = 10/5 + 2/5 = 12/5$
- Double-Check with last term: $4 = 12/5 + 8/5 = 20/5$

Case 3 Example

So we have

$$\hat{y}(s) = \frac{2(s+2)}{(s+1)(s^2+4)} = \frac{2}{5}\left(\frac{1}{s+1} - \frac{s-6}{s^2+4}\right)$$
$$= \frac{2}{5}\left(\frac{1}{s+1} - \frac{s}{s^2+4} + \frac{6}{s^2+4}\right)$$
$$y(t) = \frac{2}{5}\left(e^{-t} - \cos(2t) + 3\sin(2t)\right)$$



Case 3 Example: Inverse Laplace Transform Again

Now consider the solution to **A Different Numerical Example:**

$$\frac{3}{s(s^2+2s+5)} = \frac{3/5}{s} - \frac{3}{5}\frac{s+2}{s^2+2s+5}$$

What to do with term:

$$\frac{s+2}{s^2+2s+5}?$$

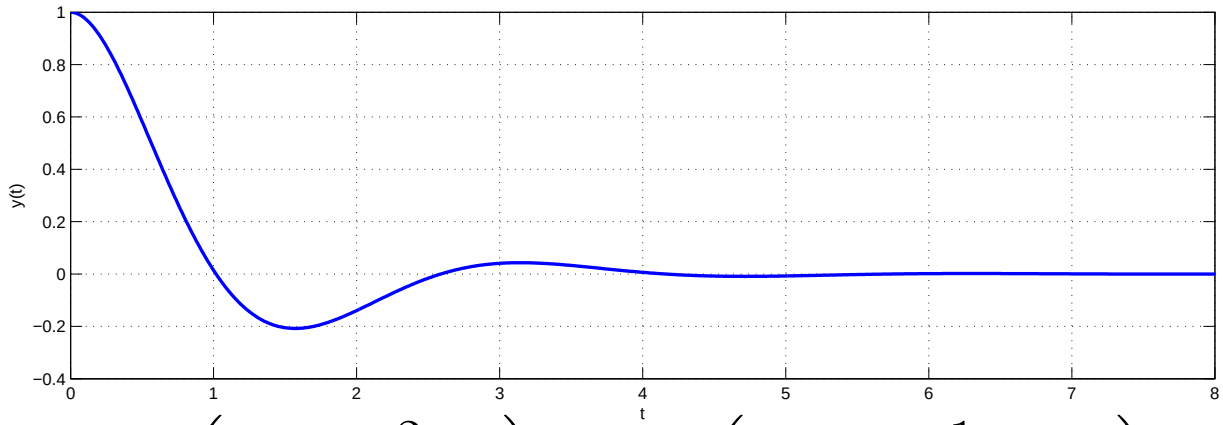
We can rewrite as the combination of a frequency shift and a sinusoid:

$$\frac{s+2}{s^2+2s+5} = \frac{s+2}{s^2+2s+1+4} = \frac{(s+1)+1}{(s+1)^2+4}$$
$$= \frac{(s+1)}{(s+1)^2+4} + \frac{1}{2}\frac{2}{(s+1)^2+4}$$

Case 3 Example: Inverse Laplace by Frequency Shifting

$$\frac{s+2}{s^2+2s+5} = \frac{(s+1)}{(s+1)^2+4} + \frac{1}{2}\frac{2}{(s+1)^2+4}$$

- $\frac{(s+1)}{(s+1)^2+4}$ is the sinusoid $\frac{s}{s^2+4}$ shifted by $s \to s+1$.
- $s \to s+1$ means multiplication by e^{-t} in the time-domain.
- $$\mathcal{L}^{-1}\left(\frac{(s+1)}{(s+1)^2+4}\right) = e^{-t}\mathcal{L}^{-1}\left(\frac{s}{s^2+4}\right) = e^{-t}\cos 2t$$
- Likewise $\mathcal{L}^{-1}\left(\frac{2}{(s+1)^2+4}\right) = e^{-t}\sin 2t$



$$\mathcal{L}^{-1}\left(\frac{s+2}{s^2+2s+5}\right) = e^{-t}\left(\cos 2t + \frac{1}{2}\sin 2t\right)$$

Summary

What have we learned today?

In this Lecture, you will learn: The Inverse Laplace Transform

- Simple Forms

The Partial Fraction Expansion

- How poles relate to dominant modes
- Expansion using single poles
- Repeated Poles
- Complex Pairs of Poles
  - Inverse Laplace

Next Lecture: Important Properties of the Response